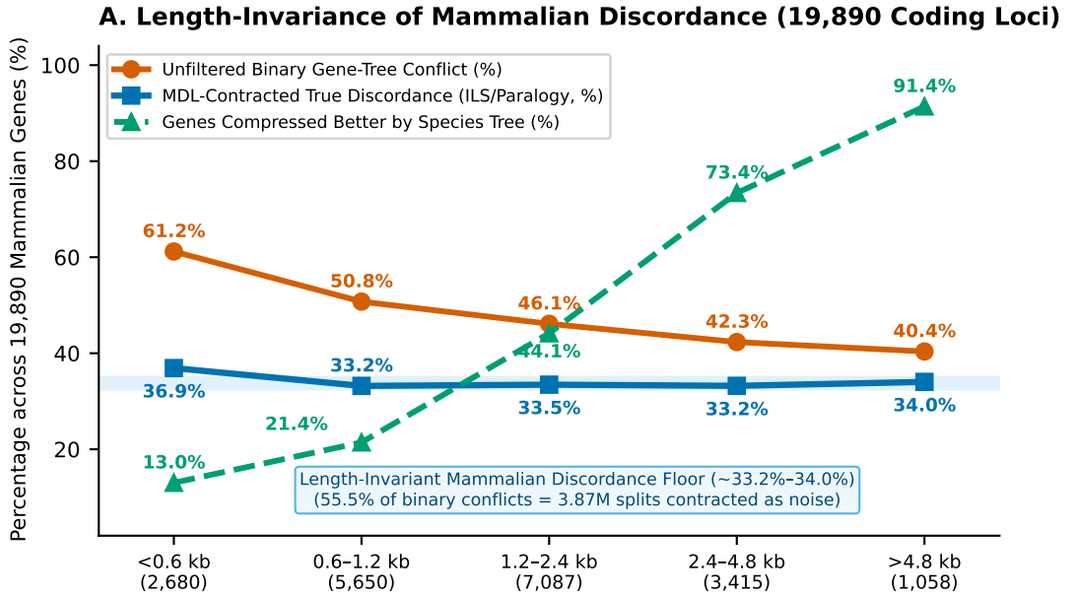
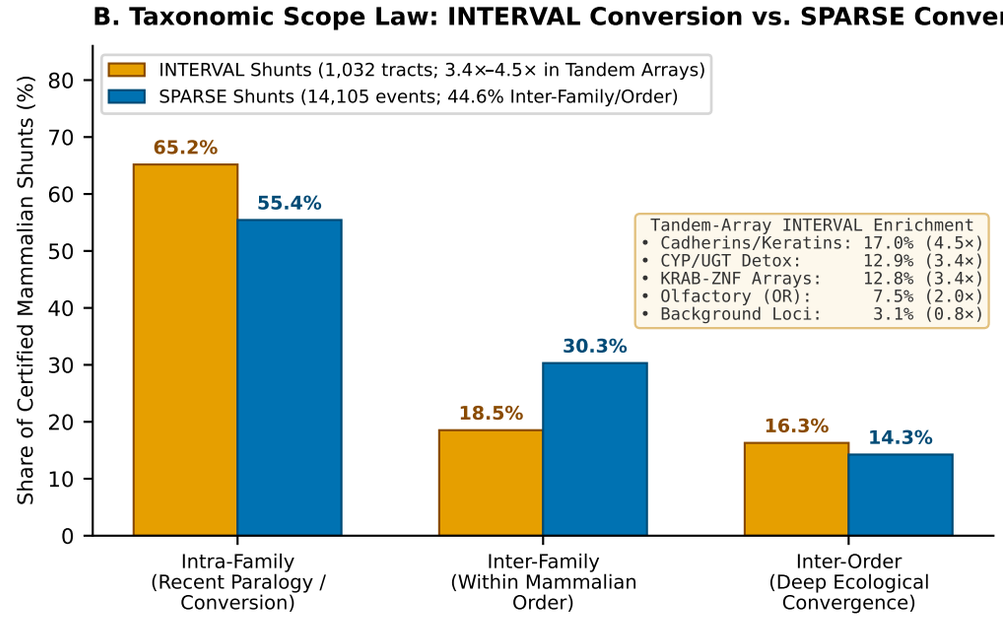
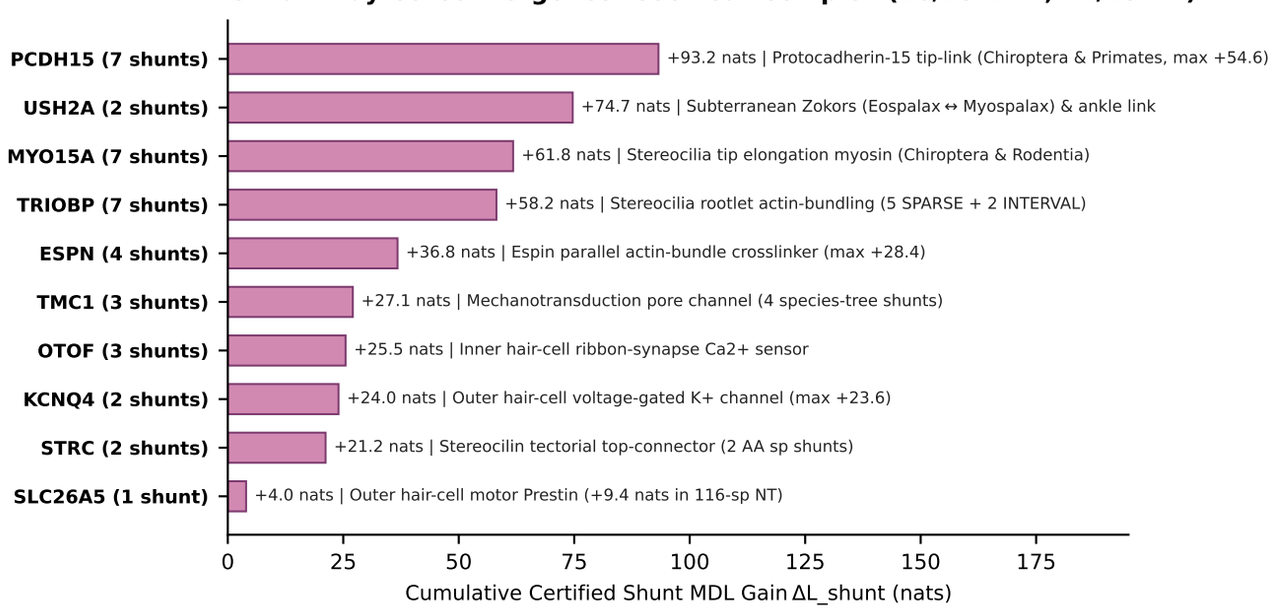
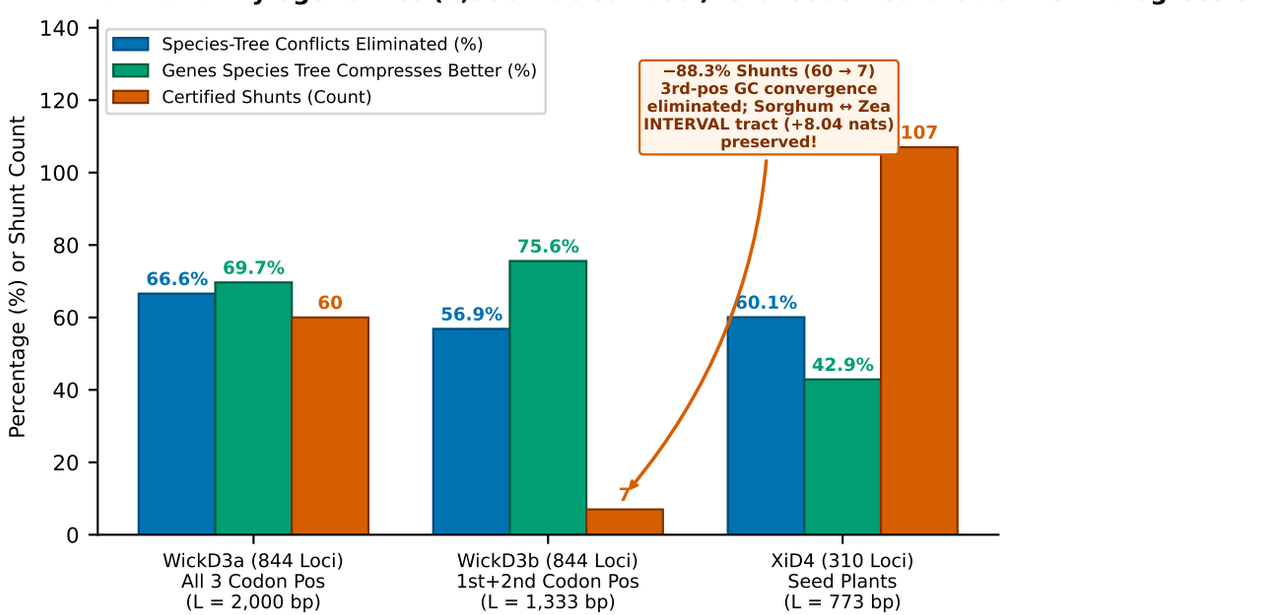

A. Length-Invariance of Mammalian Discordance (19,890 Coding Loci)
Unfiltered Binary Gene-Tree Conflict (%)
MDL-Contracted True Discordance (ILS/Paralogy, %)
Genes Compressed Better by Species Tree (%)
Percentage across 19,890 Mammalian Genes (%)
61.2%
50.8%
46.1%
42.3%
40.4%
36.9%
33.2%
33.5%
33.2%
34.0%
13.0%
21.4%
44.1%
73.4%
91.4%
Length-Invariant Mammalian Discordance Floor (~33.2%–34.0%)
(55.5% of binary conflicts = 3.87M splits contracted as noise)
<0.6 kb (2,680)
0.6–1.2 kb (5,650)
1.2–2.4 kb (7,087)
2.4–4.8 kb (3,415)
>4.8 kb (1,058)


B. Taxonomic Scope Law: INTERVAL Conversion vs. SPARSE Convergence
INTERVAL Shunts (1,032 tracts; 3.4×–4.5× in Tandem Arrays)
SPARSE Shunts (14,105 events; 44.6% Inter-Family/Order)
Share of Certified Mammalian Shunts (%)
65.2%
55.4%
18.5%
30.3%
16.3%
14.3%
Tandem-Array INTERVAL Enrichment
• Cadherins/Keratins: 17.0% (4.5×)
• CYP/UGT Detox: 12.9% (3.4×)
• KRAB-ZNF Arrays: 12.8% (3.4×)
• Olfactory (OR): 7.5% (2.0×)
• Background Loci: 3.1% (0.8×)
Intra-Family (Recent Paralogy / Conversion)
Inter-Family (Within Mammalian Order)
Inter-Order (Deep Ecological Convergence)


C. Pathway Co-Convergence: Cochlear Complex (16/25 DNA; 12/25 AA)
PCDH15 (7 shunts)
+93.2 nats | Protocadherin-15 tip-link (Chiroptera & Primates, max +54.6)
USH2A (2 shunts)
+74.7 nats | Subterranean Zokors (Eospalax ↔ Myospalax) & ankle link
MYO15A (7 shunts)
+61.8 nats | Stereocilia tip elongation myosin (Chiroptera & Rodentia)
TRIOBP (7 shunts)
+58.2 nats | Stereocilia rootlet actin-bundling (5 SPARSE + 2 INTERVAL)
ESPN (4 shunts)
+36.8 nats | Espin parallel actin-bundle crosslinker (max +28.4)
TMC1 (3 shunts)
+27.1 nats | Mechanotransduction pore channel (4 species-tree shunts)
OTOF (3 shunts)
+25.5 nats | Inner hair-cell ribbon-synapse Ca2+ sensor
KCNQ4 (2 shunts)
+24.0 nats | Outer hair-cell voltage-gated K+ channel (max +23.6)
STRC (2 shunts)
+21.2 nats | Stereocilin tectorial top-connector (2 AA sp shunts)
SLC26A5 (1 shunt)
+4.0 nats | Outer hair-cell motor Prestin (+9.4 nats in 116-sp NT)
Cumulative Certified Shunt MDL Gain ΔL_shunt (nats)


D. Plant Phylogenomics (1,998 Nuclear Loci): 3rd-Codon Saturation vs. Introgression
Species-Tree Conflicts Eliminated (%)
Genes Species Tree Compresses Better (%)
Certified Shunts (Count)
Percentage (%) or Shunt Count
66.6%
69.7%
60
56.9%
75.6%
7
60.1%
42.9%
107
−88.3% Shunts (60 → 7)
3rd-pos GC convergence eliminated; Sorghum ↔ Zea INTERVAL tract (+8.04 nats) preserved!
WickD3a (844 Loci) All 3 Codon Pos (L = 2,000 bp)
WickD3b (844 Loci) 1st+2nd Codon Pos (L = 1,333 bp)
XiD4 (310 Loci) Seed Plants (L = 773 bp)